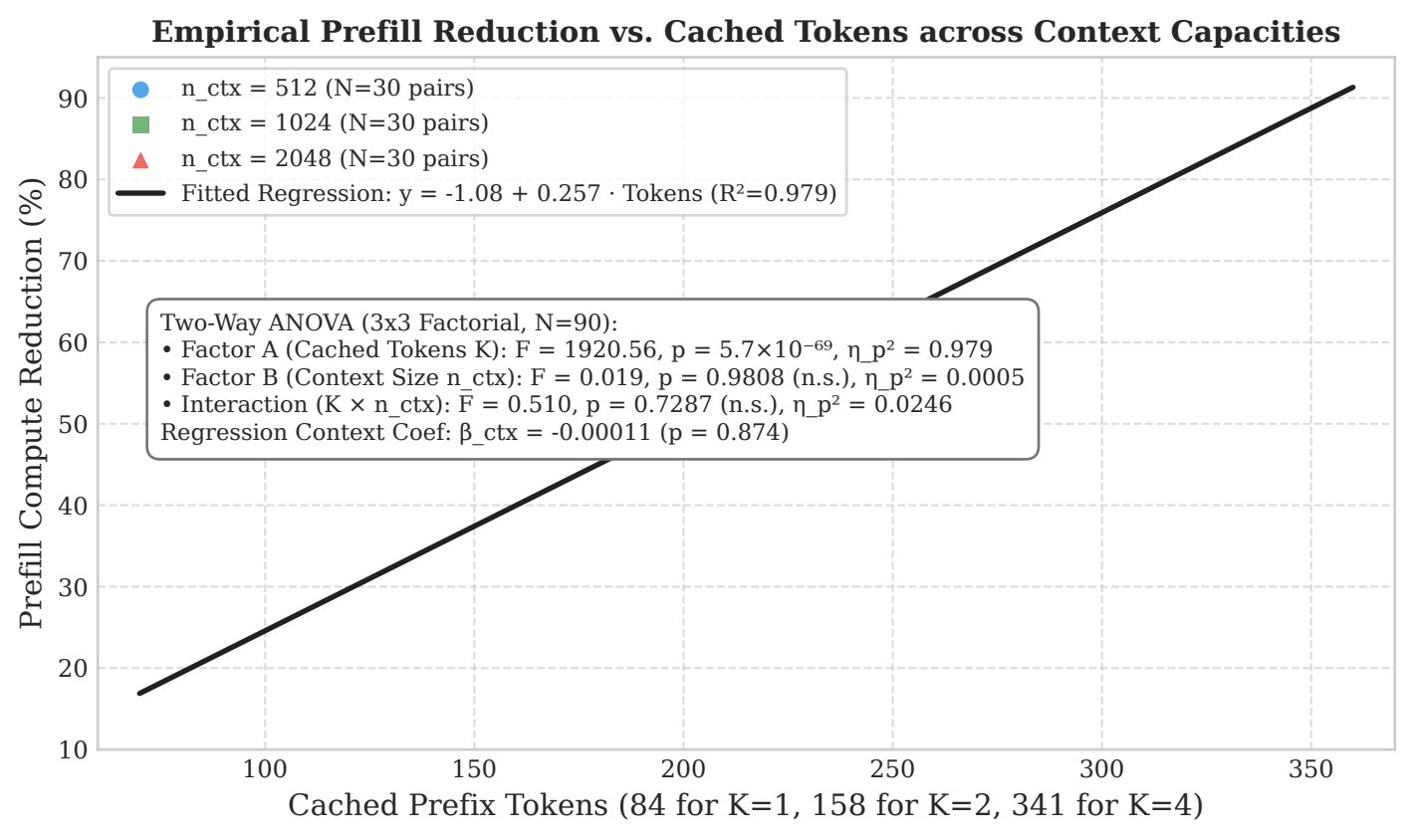
Empirical Prefill Reduction vs. Cached Tokens across Context Capacities
n_ctx = 512 (N=30 pairs)
n_ctx = 1024 (N=30 pairs)
n_ctx = 2048 (N=30 pairs)
Fitted Regression: y = -1.08 + 0.257 · Tokens (R²=0.979)
Two-Way ANOVA (3x3 Factorial, N=90):
• Factor A (Cached Tokens K): F = 1920.56, p = 5.7×10⁻⁶⁹, η_p² = 0.979
• Factor B (Context Size n_ctx): F = 0.019, p = 0.9808 (n.s.), η_p² = 0.0005
• Interaction (K × n_ctx): F = 0.510, p = 0.7287 (n.s.), η_p² = 0.0246
Regression Context Coef: β_ctx = -0.00011 (p = 0.874)
Prefill Compute Reduction (%)
10
20
30
40
50
60
70
80
90
100
150
200
250
300
350
Cached Prefix Tokens (84 for K=1, 158 for K=2, 341 for K=4)